ROHIT NALE

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PROFESSIONAL SUMMARY

Full Stack Engineer with 3+ years of experience building high-scale web applications, RESTful APIs, and enterprise data analytics platforms using Python, Django, React, TypeScript, and SQL databases. Demonstrated success in delivering data-driven software solutions, automated workflows, and interactive dashboards for large-scale enterprise and government initiatives. Experienced in end-to-end software architecture, database optimization, and Docker containerization, with a strong background in analytical data processing and cross-functional stakeholder collaboration.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (Django, FastAPI), TypeScript, React.js, Node.js, Rust (Tauri 2.0), Shell Scripting (Bash, PowerShell)
- **Databases & Data Engineering:** PostgreSQL, MongoDB, SQL Data Modeling, ETL/ELT Pipelines, Spatial & Time-Series Data Analysis
- **System Architecture & Tools:** RESTful APIs, Microservices, Docker, Git, CI/CD Pipelines, Linux/Unix
- **Analytical & ML Libraries:** PyTorch, Scikit-learn, Pandas, NumPy
- **Geospatial & Specialized Software:** QGIS, GCP, Google Earth Engin

WORK EXPERIENCE

CSIR-NEERI — *Project Associate*

Nagpur, India | Nov 2022 – Present

Enterprise Water Analytics Dashboard (Jal Jeevan Mission)

- Engineered a full-stack IoT and analytical monitoring platform for the Ministry of Jal Shakti to evaluate real-time water quality and supply metrics across national networks.
- Built scalable backend REST APIs using Python and Django to process time-series sensor data streams from edge devices.
- Architected responsive web dashboards using React, TypeScript, and OpenLayers for spatial data visualization, handling high-frequency sensor payloads.
- Optimized queries and schema design in MongoDB and PostgreSQL, reducing latency for complex analytical reporting queries.

PipeSelect Decision Support Application (CPHEEO)

- Designed and deployed PipeSelect v1.0, an enterprise decision-support tool helping municipal engineering authorities select optimal pipe infrastructure based on financial and technical criteria.
- Developed backend business logic and RESTful endpoints in Python/Django, integrating relational databases (PostgreSQL) for complex criteria evaluation.
- Delivered a desktop distribution using Tauri and web frontend interfaces, ensuring reliable execution across heterogeneous client environments.

Godavari River Basin Environmental & Hydrology Platform

- Built web-based GIS and analytical applications using Python, QGIS, and custom data processing pipelines to support the Condition Assessment and Management Plan.
- Executed hydrology modeling and automated data transformation routines using Python scripts, processing spatial and temporal environmental datasets.

- Collaborated directly with government and municipal stakeholders to translate complex domain requirements into interactive reporting dashboards and technical specifications.

Freelance — *Software & Design Consultant*
Jan 2021 – Oct 2022

- Partnered with enterprise clients to design digital branding assets, technical presentations, and custom web interfaces across multiple product domain areas.

PUBLICATION & RESEARCH

Suspicious Human Activity Detection Using Pose Estimation and LSTM
International Symposium of Asian Control Association on Intelligent Robotics (IRIA, 2021)

- Authored research paper on deep learning architectures for real-time video surveillance and activity classification using PyTorch, RNN/LSTM networks, and geometric features.
- Reached 89.07% cross-view classification accuracy on the NTU-60 dataset through spatial-temporal data feature engineering.

HACKATHON

Winner, Hackathon on R&D 2026
Organized by CMPDI, Coal India & Ministry of Coal

- Developed an advanced AI/ML-driven GIS model for automated tree enumeration, species identification, and height estimation to support sustainable land-use planning and environmental monitoring.
- Engineered end-to-end spatial processing pipelines integrating high-resolution satellite imagery and remote sensing libraries to automate canopy delineation and feature extraction.

EDUCATION

Master of Technology (M.Tech) in Communication System Engineering
Visvesvaraya National Institute of Technology (VNIT), Nagpur | 2020

Bachelor of Technology (B.Tech) in Electronics & Telecommunication Engineering
Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T), Nanded | 2018

CERTIFICATIONS

- Applied Data Science with Python — University of Michigan (Coursera)
- Deep Learning Specialization — deeplearning.ai (Coursera)
- Python 3 Programming Specialization — University of Michigan (Coursera)